Why the brass build works

Ten short chapters on the physics and engineering behind the rules in the Brass Wiring page: why two rails, why a resistor here and a capacitor there, why one corner stays empty, why two irons' worth of temperature. Written for a software engineer; each chapter has one picture and one analogy. About forty minutes, no bench needed.

Chapters

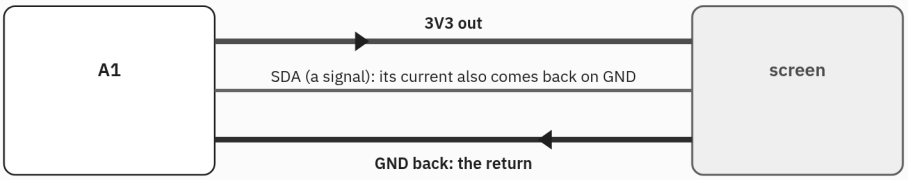
1. Everything is a loop
2. Volts, amps, ohms
3. Two voltages, one source
4. Pulls and defaults
5. Capacitors: the local cache
6. I²C: the slow bus
7. I²S: the fast bus, and frequency
8. The amplifier and the speaker
9. The antenna and the near field
10. Heat, solder and flux

Every chapter ends with the rules it explains, so you can read it either way: concept first, or start from a rule you were wondering about.

CHAPTER 1 OF 10

Everything is a loop

> **Analogy:** every call needs a return. A wire with no ground path is a function that never returns: nothing happens, and nothing tells you why.



No return path, no current, no signal. Every wire in the device is half of a loop whose other half is the ground rail.

Electricity only moves in closed loops. That is the whole reason the device has one ground rail that every module touches: it is the shared return half of every loop, power and signal alike.

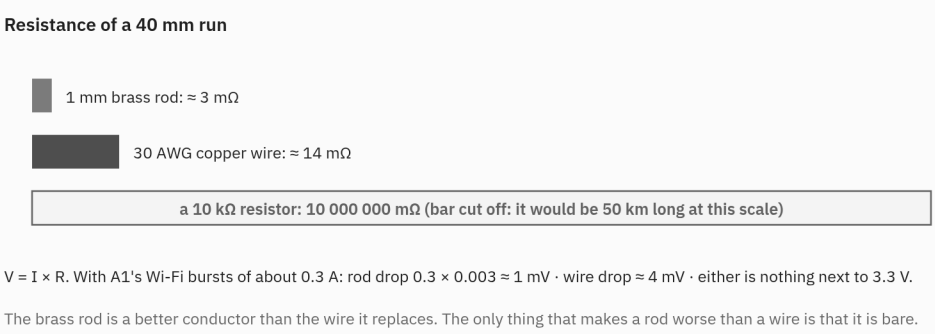
IN THE BRASS BUILD, THIS IS WHY

- one GND rail that every module taps
- the thick ground rod for the amp: the biggest return current gets its own path
- the meter check after each rod: a rail touching the frame makes the frame part of a loop

CHAPTER 2 OF 10

Volts, amps, ohms

> **Analogy:** voltage is pressure, current is flow, resistance is pipe width. The rods are wide pipes; the resistors are deliberate pinholes.



Ohm's law, V = I × R, with this device's real numbers. Conductors are near zero, resistors are deliberately huge, and the drop across a rod is a thousandth of a volt. Rods are not a compromise electrically; they only need to be kept apart.

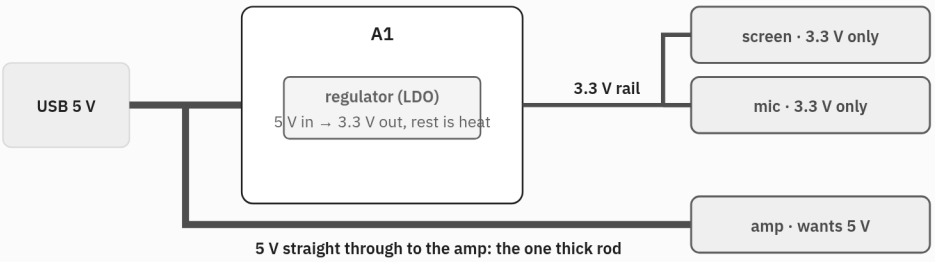
IN THE BRASS BUILD, THIS IS WHY

- bare rod is an excellent conductor; the spacing rules are the whole cost
- resistor values are chosen for how much current they *leak* (V/R), not for the signal

CHAPTER 3 OF 10

Two voltages, one source

> **Analogy:** 3.3 V and 5 V are two incompatible API versions. The regulator is the adapter; the amp speaks the old one natively.



Two voltages, one source. The screen and mic chips are rated for 3.3 V and would be damaged at 5 V; the amplifier can swing a bigger signal from 5 V, which is the volume you heard. Hence the rule: 5 V goes to the amp and nowhere else.

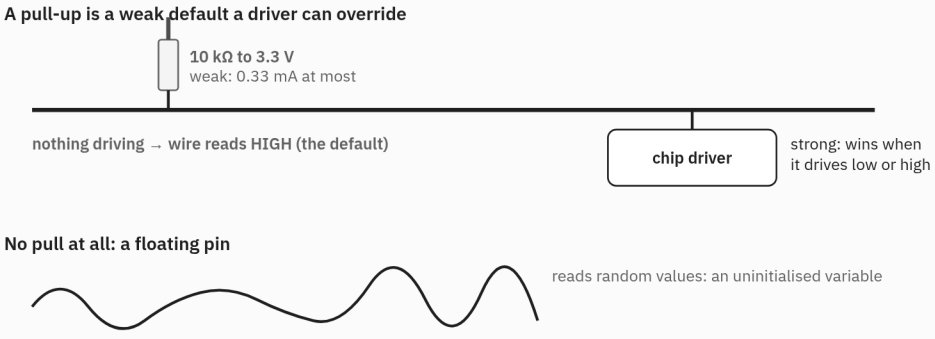
IN THE BRASS BUILD, THIS IS WHY

- the 3V3 rail feeds the screen and mic; the single 5 V rod feeds the amp
- the frame isn't powered, so it belongs to neither rail
- one USB source means there is never a moment when one part is on and another off

CHAPTER 4 OF 10

Pulls and defaults

Analogy: a pull resistor is a default argument. The caller (the chip driver) can pass a value; if it doesn't, the default applies.



A digital input is a voltage measurement, and an unconnected pin measures noise. Pull resistors give every input a default. The three in this build: 10 kΩ holds GPIO2 high because A1 reads it during reset to choose how to boot; 100 kΩ keeps the mic's data line from floating between frames; the amp's SD pull-down keeps the amp off (below 0.08 V) until A1 is awake and drives it. The button uses A1's built-in pull-up.

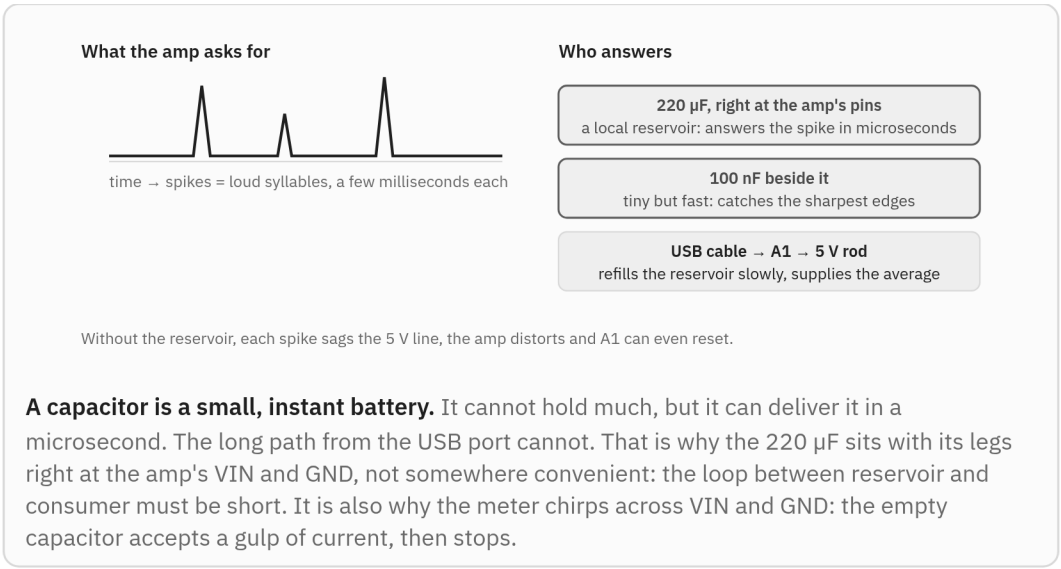
IN THE BRASS BUILD, THIS IS WHY

- 10 kΩ on GPIO2 (boot pin) to 3V3 rail
- 100 kΩ on the mic's data line to GND rail
- the amp's SD pull-down to GND rail: the amp stays off through reset
- button: A1's internal pull-up, so pressed = pulled to ground

CHAPTER 5 OF 10

Capacitors: the local cache

Analogy: the 220 µF is an L1 cache next to the CPU; the USB cable is main memory. Loud syllables are cache hits or they are stalls.



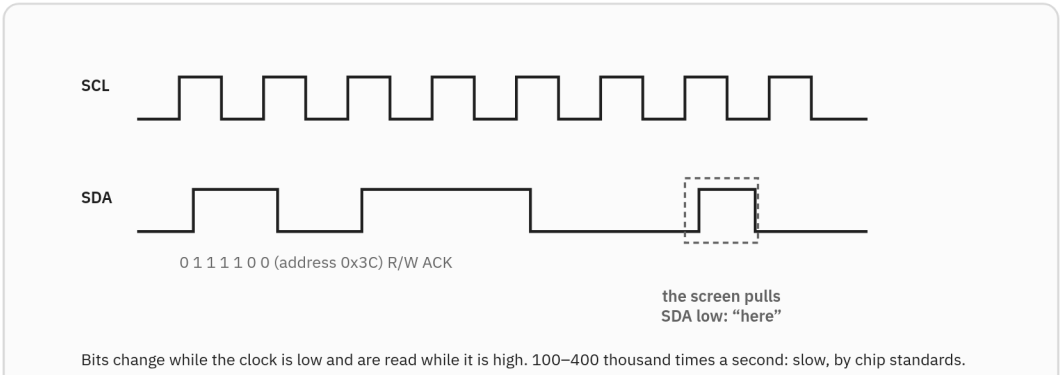
A capacitor is a small, instant battery. It cannot hold much, but it can deliver it in a microsecond. The long path from the USB port cannot. That is why the 220 µF sits with its legs right at the amp's VIN and GND, not somewhere convenient: the loop between reservoir and consumer must be short. It is also why the meter chirps across VIN and GND: the empty capacitor accepts a gulp of current, then stops.

> IN THE BRASS BUILD, THIS IS WHY
>
> - 220 µF + 100 nF soldered with the shortest possible leads at the amp's VIN and GND
> - why the meter chirps, then stops, between VIN and GND

CHAPTER 6 OF 10

I²C: the slow bus

> **Analogy:** a shared bus with device addresses, master-initiated, acknowledged per byte. Think a tiny, polite RPC over two wires.



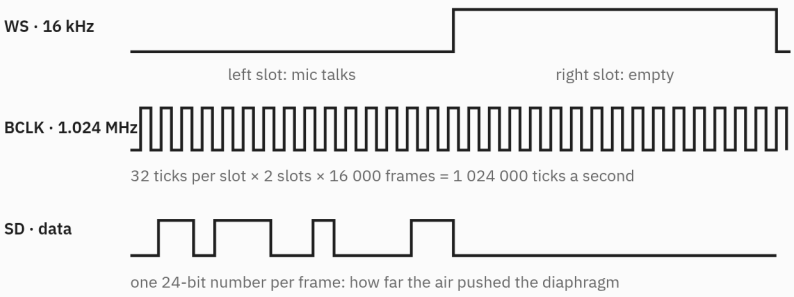
I²C is a two-wire bus with addresses, like a tiny network. A1 is the only talker; it clocks SCL and puts the address 0x3C (or 0x3D) on SDA; the screen answers. Nobody drives the lines high: devices only pull them low, and resistors on the screen module pull them up (chapter 4 again). At a few hundred kilohertz, 30 mm of brass rod is electrically a perfect wire, so the rods can run wherever is tidy.

> IN THE BRASS BUILD, THIS IS WHY
>
> - two rods, SCL and SDA, to the screen, routed wherever is tidy
> - the pull-ups live on the screen module, so nothing extra is needed

CHAPTER 7 OF 10

I²S: the fast bus, and frequency

> **Analogy:** a fixed-rate stream with a frame clock and a bit clock, like a PCM audio buffer on the wire.



16 000 samples a second captures sound up to 8 kHz, which covers speech.
The mic and the amp share WS and BCLK: one rod each, two taps.

I²S is the same idea at a thousand times the speed, with a clock the firmware chose. A1 generates both clocks, so sharing them between mic and amp is natural: that is why those two rods have two taps. The mic's L/R pin wired to GND makes it speak in the left slot, the one the firmware reads; get that wrong and you see samples counting with min and max stuck at 0. At 1 MHz the wavelength is 300 m, so a 40 mm rod is still electrically short, but this is the fastest signal in the box, so it is the one to keep away from the antenna corner.

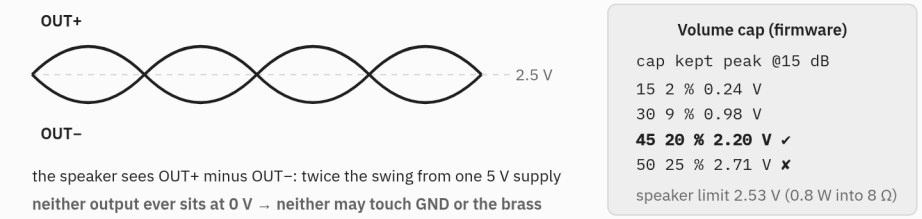
IN THE BRASS BUILD, THIS IS WHY

- WS and BCLK rods each have two taps: mic and amp share the clocks
- the mic's L/R link to GND picks the left slot
- these are the rods to keep away from the antenna corner

CHAPTER 8 OF 10

The amplifier and the speaker

> **Analogy:** the bridge output is a differential signal; the speaker reads the difference, like a differential pair, so there is no ground reference to borrow.



Class-D and the bridge. The amp switches its outputs at a few hundred kHz and the speaker's coil smooths that into sound. Driving the speaker between two outputs doubles the swing, which is why a 5 V part is loud, and why both speaker leads are live. The firmware squares the cap before applying it, so cap 45 passes only 20 % of full amplitude: 2.20 V peak even at the 15 dB gain your 100 kΩ GAIN resistor selects, under the speaker's 2.53 V rating.

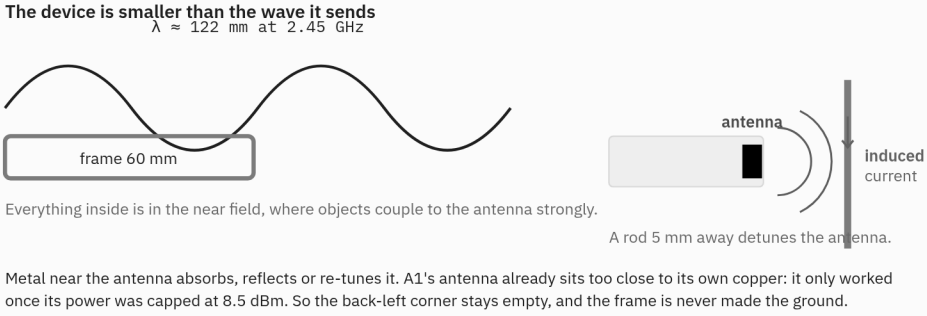
IN THE BRASS BUILD, THIS IS WHY

- OUT+ and OUT− never touch GND, a rail or the frame
- the 100 kΩ GAIN resistor to the GND rail = 15 dB
- the volume cap stays at 45 unless the scope measures a higher one

CHAPTER 9 OF 10

The antenna and the near field

Analogy: nearby metal is a side effect you didn't declare: it changes the antenna's behaviour without any explicit connection.



Why a corner of the box stays empty. At 2.4 GHz the antenna's field fills the whole frame, and any conductor in it becomes part of the antenna. Grounding the frame would wrap the antenna in a ground plane; running a rail past it would add a parasitic element. The last step of the build, comparing Wi-Fi inside the frame with the open bench, is the only real test; the clearance rules just give it the best chance.

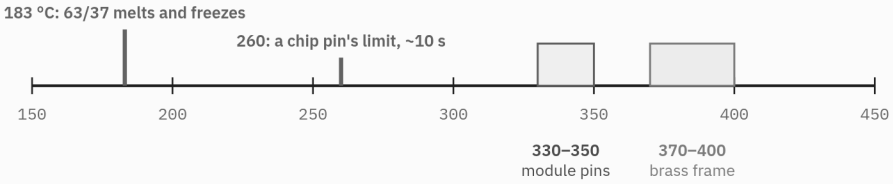
IN THE BRASS BUILD, THIS IS WHY

- the back-left corner stays empty with nothing above it
- the frame is never ground
- the fast I²S rods stay on the other side of the box

CHAPTER 10 OF 10

Heat, solder and flux

> **Analogy:** two environments with separate tooling. Brass tip and acid flux are prod-only ops tools; they never run against the dev boards.



The iron runs far above the solder's melting point because the joint must reach 183 °C quickly, before heat soaks into the part. Brass is a thick heat sink, so it gets the hotter tip and the hotter setting; a module pin gets 1–2 seconds and no more.

Soldering is a race between the joint and the part. 63/37 is a eutectic alloy: it goes from solid to liquid at one temperature, 183 °C, with no pasty range, so joints that cool undisturbed come out shiny. Flux strips the oxide that stops solder wetting metal. The Harris flux does that with an acid strong enough for brass, and its residue conducts and corrodes, which is why it never comes near a module: a few microamps of leakage on an I²S line is a fault you would never find.

IN THE BRASS BUILD, THIS IS WHY

- brass joints: 370–400 °C, acid flux, then wash
- module joints: 330–350 °C, Chip Quik flux, 1–2 s, electronics tip
- wash the frame before any part comes near it

Check yourself

Why does the meter chirp between VIN and GND, then go quiet?

The empty 220 µF capacitor takes a gulp of current and looks like a short for a moment; once charged it stops. A steady tone would be a real short.

Why can't I just ground the brass frame like the video?

Both speaker outputs are live, the 5 V rod is live, and a grounded cage next to the antenna acts as a ground plane. One accidental touch becomes a short, and Wi-Fi gets worse.

Why is a 1 mm brass rod fine for a 1 MHz clock?

At 1 MHz the wavelength is 300 m; a 40 mm rod is electrically a dot. It only matters that it stays clear of other rods and the antenna.

Why 10 kΩ and not 100 Ω for the pull-up?

The chip's driver must be able to override it, and a 100 Ω pull-up would waste 33 mA whenever the line is low. 10 kΩ wastes 0.33 mA and still wins against noise.

Why does the frame get a hotter iron than the chips?

Brass is a big heat sink; the joint must reach 183 °C fast. A chip pin can take about 260 °C for ten seconds, so module joints are quick and cooler.

Parts mentioned

PRODUCT YOU BOUGHT	IN THIS READING	LINK
K&S #9861 1.0 mm brass rod	chapters 2, 9	B005WPB7YG
LuminologyPro resistor kit	chapter 4, 8 (10 kΩ, 100 kΩ, pull-down, GAIN)	B0F4P352BB
ALLECIN electrolytic kit (220 µF) and BOJACK ceramic kit (100 nF)	chapter 5	owned · no link recorded
Hosyond SSD1306 OLED	chapter 6	B09T6SJBV5
AITRIP INMP441 microphone	chapter 7	B092HWW4RS
HiLetgo MAX98357A amplifier · Same Sky CES-20134-088PM speaker	chapter 8	B0CDWXZZCH · DigiKey
Meshnology ESP32-C3 SuperMini	chapters 3, 4, 9	B0F888JQ91
MAIYUM 63/37 solder · Harris SCLF4 flux · Chip Quik CQ4LF pen	chapter 10	B076QF1Y85 · B0015DWPV8 · owned

Read deeper

- Short notes: `plan/concepts/` 01–04 (charge, Ohm, series/parallel, capacitors), 08 (I^2C), 09 (I^2S), 10 (amplifiers), 13 (power integrity), 14 (fit and radio).
- Full lessons: `edu/fundamentals/` 06 (power integrity and decoupling), 07 (GPIO and boot straps), 08, 09, 10, 11 (RF), 12 (soldering).
- The illustrated pages: Fit and Radio and Fast Track Theory.

Sources. The device-specific numbers come from the project: the 1.024 MHz bit clock, the 0.08 V shutdown level, the squared volume cap and the 2.53 V speaker limit from `plan/FAST_TRACK.md` and `plan/FAST_TRACK_THEORY.md`; the 8.5 dBm cap and antenna notes from `docs/INVENTORY.md`. Brass resistivity about 6.4×10^{-8} Ω·m and 30 AWG copper at 0.34 Ω/m are handbook values; 183 °C is the 63/37 eutectic point; the 260 °C / 10 s figure is a typical soldering rating for a through-hole part, not from your parts' datasheets.

What is and is not verified. This is theory tied to your build's numbers. Nothing here replaces the tests in the build pages; the Wi-Fi comparison at the end is the only proof the antenna chapter gets.